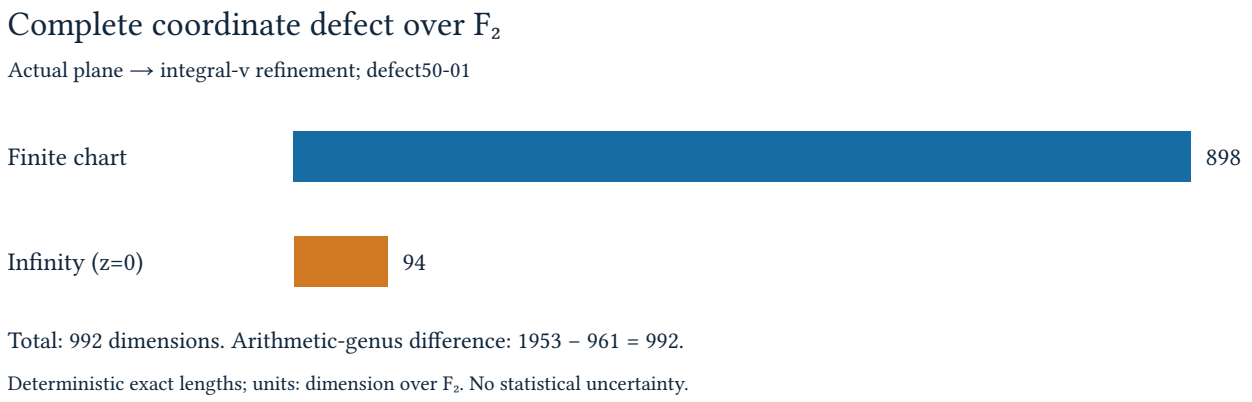
R50: explicit proper charts and exact coordinate saturation

The source field has an explicit monic degree-32 model on two affine charts. The integral-v refinement and its overlaps are checked against that field. The complete coordinate defect has dimension 992 over F_2, including a cyclic 128-dimensional boundary block with exact parameter exponent 8.



Checked construction

The quadratic norm recovers v and n rationally. Native Euclidean certificates give the common denominator inverse; all four original forward source equations satisfy full denominator-cleared polynomial products, with corrupted witnesses rejected. The scaled $t=u^4\lambda$ equation is monic of degree 32. Two monic charts and their complete overlap products construct a finite locally free proper model. The intermediate ring retains integral v with a monic degree-16 equation over $v^2+v=u^3+u$; both norm products and all twelve relative overlap products pass.

The actual u=1 fiber separates two refined degree-2 points above one plane point. A source-module presentation identifies the coordinate defect with $F_2[t, u]/(A, B)$. Its Bézout splitting isolates the cyclic block of dimension128, with $(u+1)^7$ nonzero and $(u+1)^8$ zero. Full finite and infinity lengths are898 and 94. The infinity truncations at exponents1, 2, 4, 8, 16, 32 give dimensions 16, 32, 64, 86, 94, 94, including the multiplicities. Transition weights independently give arithmetic genera1953 and 961. These are native identities and written geometric deductions; the linked notes state their hypotheses.

Formal algebra

Twelve unique mathematical declarations compile and pass the axiom audit: six Rational50 norm, coordinate, field-map and overlap lemmas; one Separation50 evaluation-separation lemma; and three Local50 scaling, constructive CRT and nilpotency lemmas;two Marked23_50 binary-prefix and prime-order lemmas. Zero failing or trusted declarations. Actual CAS arrays, scheme gluing and complete pole geometry are outside these Lean propositions. Lean4.34.1/mathlib d13f23b, one thread, 4096 MiB, 1200 s per compile/audit.

Complete RR algebra and larger marked base

All 68 descended sections and 196 complete-RR products have actual 32-entry field vectors, complete denominator identities and full reconstruction. Coefficient inversion transports them to z=1/u. The PID module has rank 32, common denominator degree 37 and index length 376 over F_2. Both original-column lifts and the entire source power-basis embedding pass.

All 528 unordered products of the 32 basis functions are checked in the actual source quotient. The complete 32-by 528 table identity and corruption control pass. A fresh process reloads the actual serialized parents, products, table and source embeddings and rechecks all complete identities. This gives an explicit finite free source algebra. The written RR and Nakayama argument identifies it with the affine normalization under the retained genus 61, proper source, pole support and complete-space hypotheses. Those hypotheses are stated in the notes and not formalized by the conditional Lean lemmas.

The known sign quotient now has monic degree 8 equations on both proper charts, full coupled-source identities, generic ranks16 and 32, and five overlap products. Its arithmetic genus is 449. It supplies source geometry for reconstructing the anti-invariant double cover.

The larger marked23 base passes over the actual prime degree 7 characteristic2 field. The Tate-coordinate inverses, nonzero discriminant, 2P/3P controls and every binary prefix 1, 2, 5, 11, 23 pass. The final point is infinity and P is nonidentity, hence has exact order23. All actual arrays are retained. Next apply only universal QR7 equations S3-S5 to this base;degree 112 remains a prospective extension value.

The next proper-source gluing uses the RR algebra and its diamond transport. The overlap begins with h0=z+V, h1=z+V+z^2;full module lifts must certify regularity of every transported generator, including exceptional branches. A complete 132-section finite-u RR intersection from saved1476-dimensional spaces is a separate route. See NEXT_SOURCE_NORMALIZATION.md.

Actual attempts and receipts

30 attempts: 20 accepted, 10 rejected.

Attempt	Status	Receipt
boundary50-01	accepted	exit 0
candidate50-01	accepted	exit 0
chart50-01	accepted	exit 0
defect50-01	accepted	exit 0
denominator50-01	rejected	exit 0
denominator50-02	accepted	exit 0
fiber50-01	accepted	exit 0
forward50-01	rejected	exit 0
forward50-02	accepted	exit 0
lean50-01	accepted	compile 0, audit 0
local-lean50-01	accepted	compile 0, audit 0
local50-01	accepted	exit 0
marked-lean50-01	rejected	compile 1, audit None
marked-lean50-02	accepted	compile 0, audit 0
marked23-50-01	rejected	exit 0
marked23-50-02	accepted	exit 0
normal-finite50-01	rejected	exit 124
normal-relative-finite50-01	rejected	exit 0
normal-relative-infinity50-01	accepted	exit 0
quotient50-01	accepted	exit 0
relative50-01	accepted	exit 0
rr-field50-01	accepted	exit 0
rr-mult50-01	rejected	exit 0
rr-mult50-02	accepted	exit 0
rr-pid50-01	accepted	exit 0
rr-table-reload50-01	accepted	exit 0
separation50-01	rejected	compile 1, audit None
separation50-02	rejected	compile 1, audit None
separation50-03	accepted	compile 0, audit 0
verify-relative-finite50-01	rejected	exit 124

Preserved failures and next construction

The unsupported coefficient-field extgcd path and direct polynomial division path fail explicitly; supported Euclid and leading-term division repair them in fresh attempts. Two narrow-import Lean compiles fail before the accepted separation module. The weighted plane normalizer reaches normalP and caps at 1200 s without module output. The refined finite normalizer exports candidate arrays but fails a coefficient-display-string equality: Singular displays the prime field as ZZ/2. That producer remains rejected. A separately frozen finite verifier hits1200 s in the 22-variable standard-basis construction. The repaired infinity producer exports a14-variable presentation with delta 376; its independent presentation verifier remains prepared, unexecuted.

The first marked23 native attempt fails a reserved identifier;the repaired run passes. Its first Lean compile misses distributivity;the repaired build and audit pass. The PID writer's comment header swallows a declaration; the first table consumer rejects. The repaired consumer strips only that comment prefix, rechecks both full saved module products, and the accepted table subsequently passes a fresh reload. Every failure remains indexed.

All native CAS invocations use4 GiB virtual memory, 1200 s, seed50 and one thread, serially. Failed late PASS markers never count. R27–R49 stay sealed. CPU time was not instrumented;actual native UTC wall timestamps are retained. No new research EKS job, Diem/freezer restart, Conductor claim or agent merge. This report describes component checks; independent scientific replay and semantic review have not been performed. See LOCAL-GEOMETRY.md, DEFECT-GEOMETRY.md, RR-NORMALIZATION-LEAD.md and the exact attempt logs.